ENDG 319

# Probability, Statistics and Machine Learning

## Lecture 13: Confidence Intervals

Large-sample intervals for a population mean

Confidence • Precision • Sample size • One-sided bounds

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# Learning outcomes for Lecture 13

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**FROM A POINT ESTIMATE TO A DEFENSIBLE UNCERTAINTY STATEMENT**

| | | |
|---|---|---|
| 1 | **Distinguish** | Separate individual variability from uncertainty in a mean. |
| 2 | **Construct** | Build a large-sample confidence interval and choose its cutoff. |
| 3 | **Interpret** | Explain confidence using repeated-sampling coverage. |
| 4 | **Plan** | Recover a confidence level and choose a target sample size. |
| 5 | **Bound** | Calculate and interpret one-sided confidence bounds. |

A complete result states the parameter, interval, confidence level, units, and assumptions.

# A point estimate does not describe its own uncertainty

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**ONE NUMBER ESTIMATES A PARAMETER; IT DOES NOT SHOW PRECISION**

### POPULATION MEAN

$$\bar{x} \longrightarrow \mu$$

Use the sample mean to estimate the population mean.

### SUCCESS PROBABILITY

$$\hat{p} \longrightarrow p$$

Use a sample proportion to estimate a success probability.

Different random samples generally produce different estimates.

Our task today: quantify uncertainty in an estimated population mean.

An interval adds a measure of precision; it does not make the point estimate equal to the truth.

# Standard deviation is not the standard error

**READ CAREFULLY: WHAT QUANTITY DOES THE STATED UNCERTAINTY DESCRIBE?**

### INDIVIDUAL OBSERVATIONS

$$\sigma \quad \text{or its estimate } s$$

Spread of individual measurements.

### SAMPLE MEAN

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Spread of sample means across repeated samples.

### ESTIMATED STANDARD ERROR

$$\widehat{\mathrm{SE}} = \frac{s}{\sqrt{n}}$$

Estimate the uncertainty in the mean when σ is unknown.

All three are in the original measurement units; they are not the same quantity.

If the problem already gives the standard error of the mean, do not divide by √n again.

# Example 1: piston measurements and a three-SE interval

**ORIGINAL INPUTS: INDEPENDENT MEASUREMENTS FROM A NORMAL POPULATION**

$$\bar{x} = 14.0 \text{ cm}$$

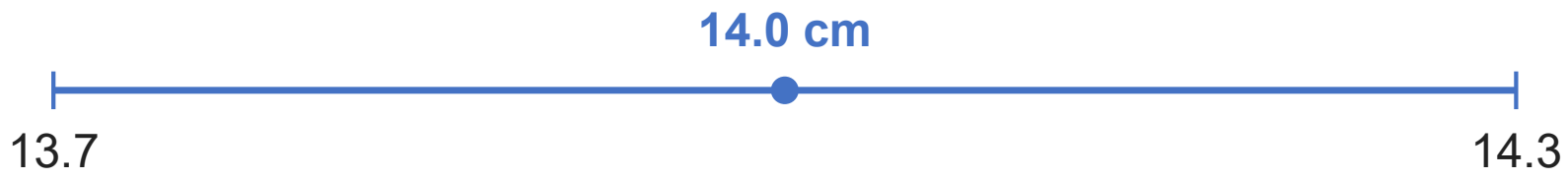
$$\sigma_{\bar{X}} = 0.1 \text{ cm}$$

The stated 0.1 cm is the uncertainty in the sample mean.

## A three-standard-error interval

$$14.0 \pm 3(0.1)$$

$$(13.7,\ 14.3) \text{ cm}$$




Under the normal sampling model, ±3 standard errors gives approximately 99.7% coverage.

# From the sampling distribution to an interval for μ

**THE SOURCE'S PISTON-DIAMETER CONSTRUCTION USES n = 100**

$$X_1, \ldots, X_{100} \quad \longrightarrow \quad \bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$$

**CENTRE OF THE SAMPLING DISTRIBUTION**

$$E[\bar{X}] = \mu$$

**SPREAD OF THE SAMPLING DISTRIBUTION**

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{100}}$$

Normal observations: the mean is normal. Otherwise: use a justified CLT approximation.

The population mean μ is unknown and fixed; the sample mean varies from sample to sample.

# Check the working conditions before calculating

**A LARGE SAMPLE DOES NOT RESCUE AN UNSUITABLE SAMPLING DESIGN**

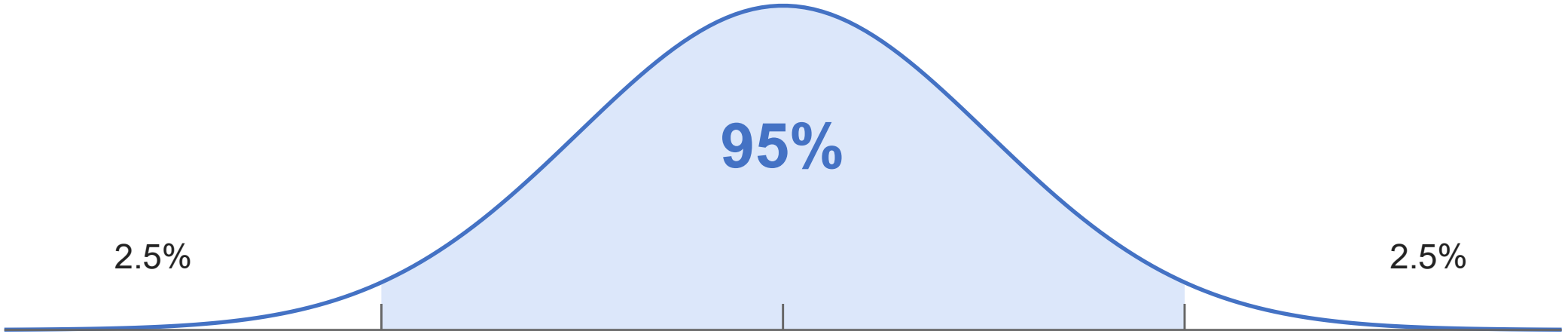
| | |
|---|---|
| **Sampling** | Use a random sample that represents the population of interest. |
| **Dependence** | The $\sigma/\sqrt{n}$ standard-error rule assumes independent observations. |
| **Shape** | The mean must be normal or adequately approximated by a normal law. |
| **Unknown σ** | Replacing σ by s gives the source's approximate large-sample z interval. |

*The source's $n > 30$ guideline is a rule of thumb, not a universal guarantee.*

Check how the data were obtained as well as the numerical sample size.

# The middle 95% describes possible sample means

**THE CURVE IS THE SAMPLING DISTRIBUTION OF THE MEAN—NOT RAW DATA**




$$\mu - 1.96\sigma_{\bar{X}} \qquad \mu \qquad \mu + 1.96\sigma_{\bar{X}}$$

$$P\left(\mu - 1.96\sigma_{\bar{X}} \leq \bar{X} \leq \mu + 1.96\sigma_{\bar{X}}\right) \approx 0.95$$

The 95% refers to repeated sample means relative to the fixed population mean μ.

# Why an interval centred on the sample mean works

THE SAME DISTANCE CAN BE READ IN TWO DIRECTIONS

## Sample mean close to the population mean

$$\left|\bar{X} - \mu\right| \leq 1.96\sigma_{\bar{X}}$$

*is equivalent to*

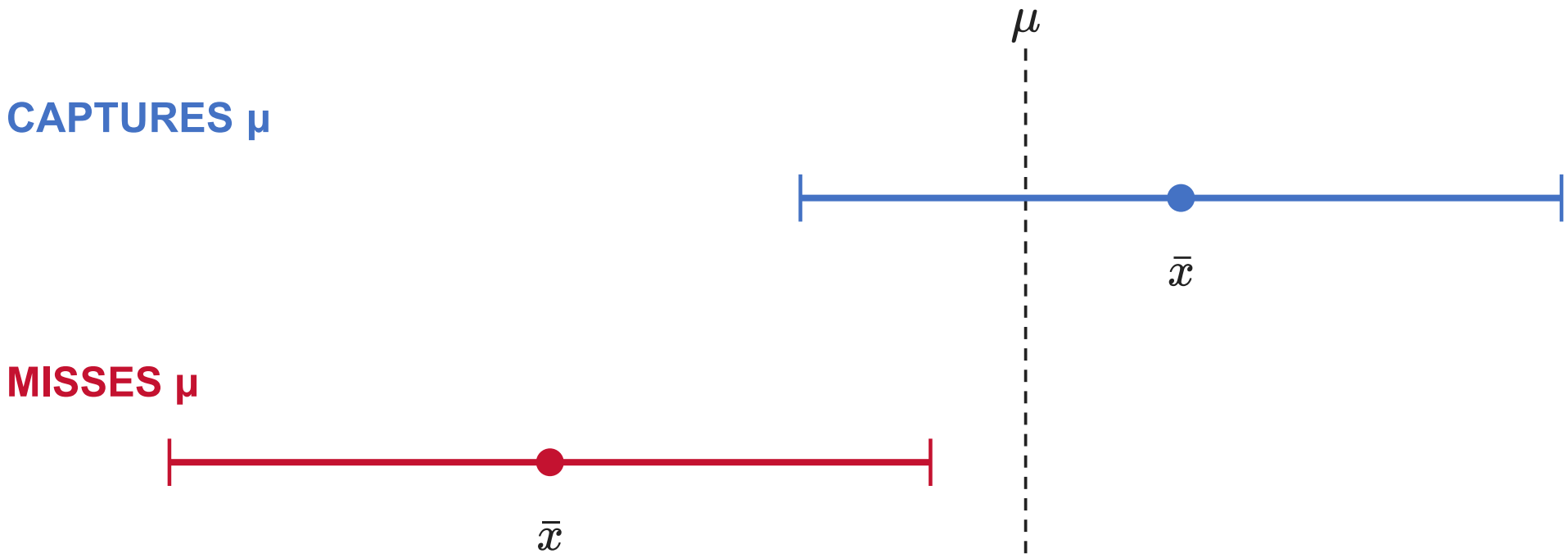
$$\bar{X} - 1.96\sigma_{\bar{X}} \leq \mu \leq \bar{X} + 1.96\sigma_{\bar{X}}$$

The sample determines the endpoints. The unknown μ is the target between them.

If $\bar{X}$ is within 1.96 standard errors of μ, its confidence interval contains μ.

# The same rule can capture μ—or miss it

**ILLUSTRATIVE POSSIBLE SAMPLES; THE POPULATION MEAN DOES NOT MOVE**




Both intervals use the same half-width: 1.96 standard errors.

Unusually low or high sample means can produce intervals that do not cover μ.

# Example 1: calculate the 95% confidence interval

**USE THE GIVEN STANDARD ERROR OF 0.1 cm DIRECTLY**

| Point estimate | Critical value | Standard error |
|---|---|---|
| 14.0 cm | 1.96 | 0.1 cm |

$$M = 1.96(0.1) = 0.196 \text{ cm}$$

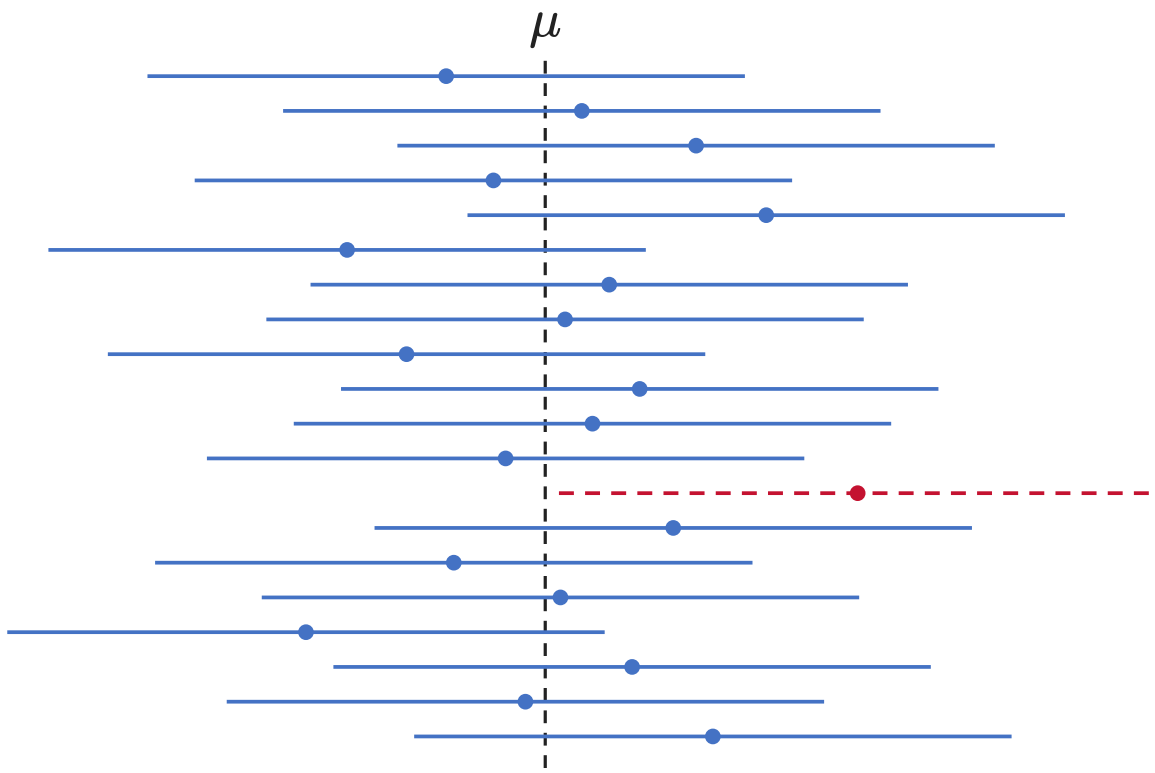
$$14.0 \pm 0.196 = (13.804,\ 14.196) \text{ cm}$$

We are 95% confident that the population mean lies between these limits.

> The interval estimates a population mean—not the range of individual piston measurements.

# Confidence describes the interval-building procedure

**REPEAT THE SAMPLING AND THE SAME CONSTRUCTION MANY TIMES**




Each row represents a fresh sample (schematic).

## Intervals move. μ does not.

About 95% of intervals cover μ in the long run when the method's conditions hold.

A short run need not achieve exactly 95%.

A 95% confidence method is reliable in repeated use; it does not guarantee any one interval.

# Random interval before sampling; fixed interval afterwards

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**KEEP FREQUENTIST CONFIDENCE DISTINCT FROM A POST-DATA PROBABILITY**

## BEFORE SAMPLING

$$P(L \leq \mu \leq U) \approx 0.95$$

L and U are functions of a random sample.
The parameter μ is fixed.

## AFTER THE SAMPLE IS OBSERVED

$$(l, u) = (13.804, 14.196)$$

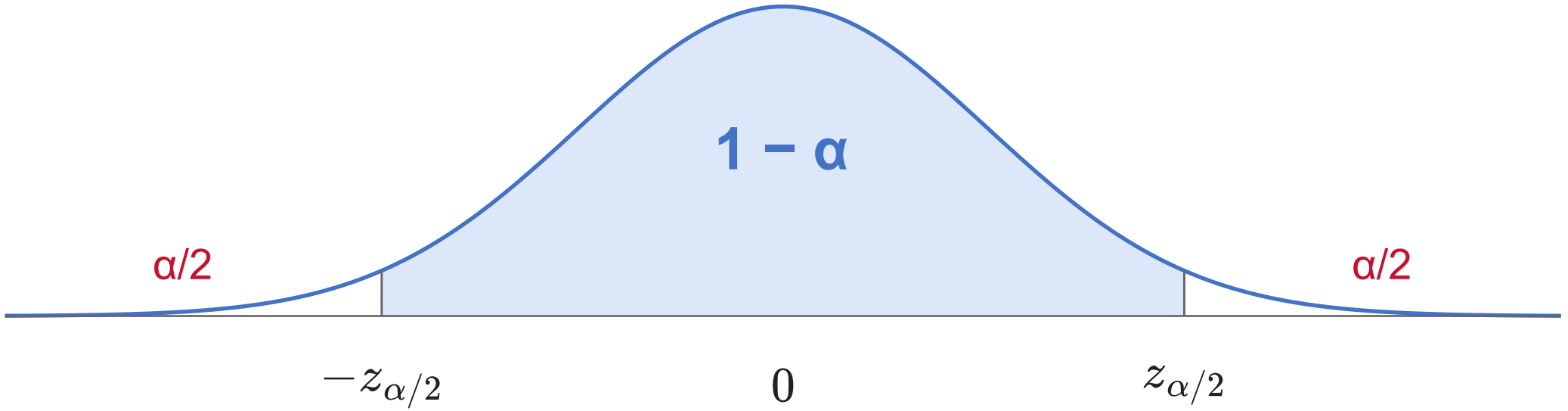
The endpoints are now fixed.
This interval either contains μ or it does not.

"95% confident" refers to the method that produced the observed interval.

# Other confidence levels: allocate the tail probability

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**USE POSITIVE UPPER-TAIL STANDARD NORMAL CUTOFFS**




$$P(Z > z_q) = q \qquad \Phi(z_{\alpha/2}) = 1 - \frac{\alpha}{2}$$

For a two-sided interval, split α equally: α/2 in each tail.

# Large-sample confidence interval for a population mean

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**POINT ESTIMATE ± CRITICAL VALUE × STANDARD ERROR**

## Population standard deviation σ known

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

## σ unknown: use s for the source's large-sample approximation

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

Replacing σ by s retains the large-sample z method; its confidence level is approximate.

# Frequently used two-sided confidence levels

THESE ARE THE ROUNDED NORMAL MULTIPLIERS USED IN THE SOURCE

| Confidence level | Positive multiplier | Margin of error |
|---|---|---|
| ≈68% | 1.000 | 1.000 × SE |
| 90% | 1.645 | 1.645 × SE |
| 95% | 1.960 | 1.960 × SE |
| 99% | 2.580 | 2.580 × SE |
| ≈99.7% | 3.000 | 3.000 × SE |

The 1-SE and 3-SE levels are approximately 68.27% and 99.73% under normality.

Do not use 1.96 automatically: match the critical value to the stated confidence level.

# A reliable five-step calculation

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**ORGANIZE THE SOLUTION BEFORE SUBSTITUTING NUMBERS**

| | | |
|---|---|---|
| **1** | **Specify** | Identify μ, the units, and the confidence level $1 - \alpha$. |
| **2** | **Select** | Two-sided interval: choose the cutoff with upper-tail area $\alpha/2$. |
| **3** | **Compute** | Find SE = $\sigma/\sqrt{n}$, or estimate it using $s/\sqrt{n}$. |
| **4** | **Construct** | Margin M = critical value × SE; limits are $\bar{X} - M$ and $\bar{X} + M$. |
| **5** | **Report** | Give the interval, confidence level, units, and sampling assumptions. |

Keep useful precision during the calculation; round the final limits sensibly.

# Example 2-1: an 85% interval for mean fill weight

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**ATTEMPT THE SETUP BEFORE MOVING TO THE WORKED SOLUTION**

The fill weights of 100 boxes have a sample mean of 12.05 ounces and a sample standard deviation s = 0.1 ounce.

**Find an 85% confidence interval for the population mean fill weight.**

What are α, α/2, the standard error, and the required positive z cutoff?

Working assumption: independent random sampling and an adequate normal approximation.

# Example 2-1: critical value, margin, and endpoints

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**WORKED SOLUTION DERIVED FROM THE SOURCE INPUTS**

$$\alpha = 0.15, \quad \alpha/2 = 0.075, \quad \Phi(z_{0.075}) = 0.925$$

$$z_{0.075} \approx 1.4395 \quad (\text{about } 1.44 \text{ from a } z \text{ table})$$

$$\widehat{\mathrm{SE}} = \frac{0.1}{\sqrt{100}} = 0.01 \text{ oz}$$

$$M = 1.4395(0.01) \approx 0.0144 \text{ oz}$$

$$85\% \text{ CI: } (12.0356,\ 12.0644) \text{ oz}$$

The target is the population mean fill weight—not a range for individual boxes.

# Example 3: recover a missing confidence level

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**WORK BACKWARDS FROM THE REPORTED INTERVAL**

A sample of 50 microdrills has mean lifetime 12.68 holes and sample standard deviation 6.83 holes.

$$\text{Reported confidence interval: } (11.09,\ 14.27)$$

## What confidence level was used?

Start by checking its centre and calculating its half-width.

Recover the multiplier using half-width ÷ standard error—not full width ÷ standard deviation.

# Example 3: the reported interval is approximately 90%

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**WORKED SOLUTION—ROUNDING OF THE ORIGINAL LIMITS MATTERS**

$$M = \frac{14.27 - 11.09}{2} = 1.59 \text{ holes}$$

$$\widehat{\text{SE}} = \frac{6.83}{\sqrt{50}} \approx 0.9659 \text{ holes}$$

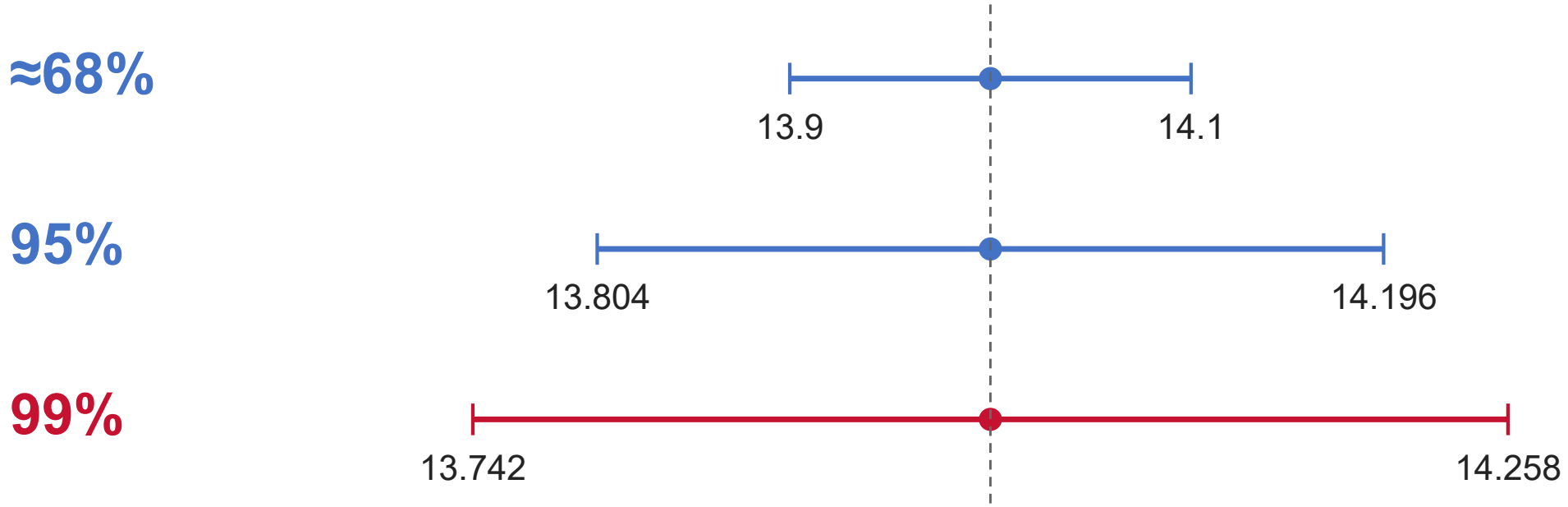
$$z^* = \frac{M}{\widehat{\text{SE}}} = \frac{1.59}{0.9659079} \approx 1.6461$$

$$1 - \alpha = 2\Phi(z^*) - 1 \approx 0.9003 \approx 90\%$$

The inferred 1.646 is consistent with the source's 90% multiplier, 1.645.

# Higher confidence means a wider interval

**HOLD THE POINT ESTIMATE AND STANDARD ERROR FIXED**




Same $\bar{X}$ = 14.0 cm and SE = 0.1 cm in every row.

With the same data, higher confidence gives a wider, less precise interval.

# Example 4: is this confidence interpretation correct?

**TRUE OR FALSE—JUSTIFY YOUR ANSWER IN WORDS**

A 90% confidence interval for the mean diameter of steel rods from an extrusion machine is (14.73, 14.91) cm.

> *"The probability that the mean diameter of rods manufactured by this process is between 14.73 and 14.91 cm is 90%."*

**Which part of the claim is the problem?**

Use the distinction between a random interval procedure and the interval already computed.

# Example 4: confidence belongs to the method

**ANSWER: FALSE IN THE FREQUENTIST INTERPRETATION USED HERE**

### A correct report

"We are 90% confident that the population mean diameter is between 14.73 and 14.91 cm."

**Why?** The endpoints and μ are fixed after sampling. The interval either covers μ or it does not; the 90% describes long-run coverage of the construction method.

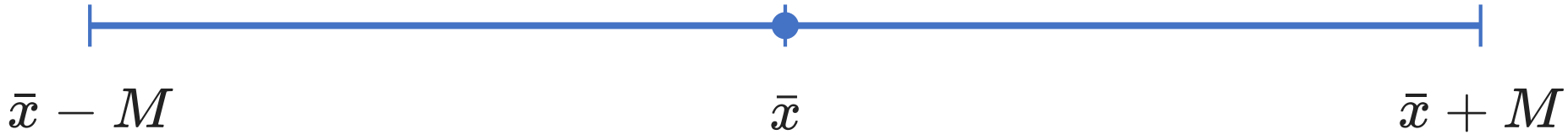
This is also not a claim that 90% of individual steel rods have diameters inside these limits.

# Margin of error is half the full interval width

**DISTINGUISH "WITHIN ±d" FROM "TOTAL WIDTH d"**

$$(\bar{x} - M,\ \bar{x} + M)$$

**Margin = M**




$$M = z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \qquad W = 2M$$

Piston example: margin = 0.196 cm; full width = 0.392 cm.

A target precision of ±0.1 cm means M = 0.1 cm and full width W = 0.2 cm.

# Choose n to meet a target margin of error

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**USE A PLANNING VALUE FOR THE POPULATION STANDARD DEVIATION**

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq d$$

$$n \geq \left(\frac{z_{\alpha/2}\sigma}{d}\right)^2$$

$$n_{\min} = \left\lceil \left(\frac{z_{\alpha/2}\sigma}{d}\right)^2 \right\rceil$$

d is the target half-width. Round up to the next whole observation.

A planning estimate of σ does not guarantee the width of a future interval.

# Piston example: plan for a 95% interval within ±0.1 cm

**SOURCE PLANNING STANDARD DEVIATION: σ ≈ 1 cm**

| Confidence | Target margin d | Planning SD |
|---|---|---|
| 95% | 0.1 cm | 1 cm |

$$n \geq \left(\frac{1.96(1)}{0.1}\right)^2 = 384.16$$

$$\boxed{n_{\min} = 385}$$

Check: n = 384 gives M ≈ 0.100021 cm; n = 385 gives M ≈ 0.099891 cm.

385 is the planned total sample size—not an additional 385 observations.

# Three levers change confidence-interval precision

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**COMPARE CHANGES WHILE HOLDING THE OTHER INPUTS FIXED**

$$M = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

| Change | Effect on the margin | Result |
|---|---|---|
| Higher confidence | Larger critical value | Wider interval |
| More variability | Larger σ | Wider interval |
| Larger sample size | Smaller σ/√n | Narrower interval |

$$\text{Half the margin} \quad \Longrightarrow \quad 4 \times \text{ the sample size}$$

Quadrupling n halves the margin; doubling n does not halve it.

# Sometimes only a lower or an upper bound is needed

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**EXAMPLE MOTIVATION: A LOWER BOUND FOR MEAN CONCRETE-BLOCK STRENGTH**

## LOWER CONFIDENCE BOUND

$$L = \bar{x} - z_\alpha \, \mathrm{SE}$$

$$[L, \ \infty)$$

Estimate how low the population mean could be.

## UPPER CONFIDENCE BOUND

$$U = \bar{x} + z_\alpha \, \mathrm{SE}$$

$$(-\infty, \ U]$$

Estimate how high the population mean could be.

A one-sided bound puts all of α in one tail; a two-sided interval splits it between two tails.

# One-sided versus two-sided: the critical value changes

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**CHOOSE THE TAIL ALLOCATION FIRST—THEN READ THE NORMAL TABLE**

| Requested statement | Tail area for cutoff | Positive cutoff |
|---|---|---|
| 95% two-sided interval | 0.05 / 2 = 0.025 | $z_{0.025} \approx 1.960$ |
| 95% one-sided bound | 0.05 | $z_{0.05} \approx 1.645$ |
| 99% one-sided bound | 0.01 | $z_{0.01} \approx 2.326$ |

$$\Phi(z_{\alpha/2}) = 1 - \alpha/2 \qquad \Phi(z_{\alpha}) = 1 - \alpha$$

For the same confidence level, a one-sided bound does not use the two-sided multiplier.

# Example 2-2: two separate one-sided requests

**RETURN TO THE SAME BOX FILL-WEIGHT DATA FROM EXAMPLE 2-1**

$$n = 100, \qquad \bar{x} = 12.05 \text{ oz}, \qquad s = 0.1 \text{ oz}$$

**A**

Find a 95% lower confidence bound for μ.

**B**

Find a 99% upper confidence bound for μ.

Which critical value and sign should be used for each request?

Treat the 95% lower bound and the 99% upper bound as separate statements.

# Example 2-2: lower and upper confidence bounds

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**WORKED SOLUTIONS DERIVED FROM THE SOURCE DATA**

$$\widehat{\mathrm{SE}} = \frac{0.1}{\sqrt{100}} = 0.01 \text{ oz}$$

## 95% LOWER BOUND

$$L = 12.05 - 1.64485(0.01)$$

$$L \approx 12.0336 \text{ oz}$$

$$[12.0336,\ \infty) \text{ oz}$$

## 99% UPPER BOUND

$$U = 12.05 + 2.32635(0.01)$$

$$U \approx 12.0733 \text{ oz}$$

$$(-\infty,\ 12.0733] \text{ oz}$$

Report each bound with its own direction, confidence level, and measurement units.

# Common mistakes—and the checks that prevent them

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**CHECK THE QUANTITY AND INTERPRETATION BEFORE THE ARITHMETIC**

| Mistake | Correction |
|---|---|
| Use s as the SE | For a mean, compute $s/\sqrt{n}$ unless SE is already given. |
| Use 1.96 for every problem | Match confidence level and number of tails. |
| Confuse margin and full width | M = half-width; W = 2M. |
| Describe individual observations | The target here is the population mean μ. |
| Assign probability to fixed limits | Confidence describes repeated-use coverage. |
| Round a required n down | Apply the ceiling to the sample-size calculation. |

Check the target, standard error, tails, and interpretation before checking the arithmetic.

# Concept check: explain before calculating

**DISCUSS BRIEFLY; THE NEXT SLIDE GIVES THE CORRECTIONS**

**A** SE = 0.1 cm is already supplied. Should it be divided by $\sqrt{100}$ again?

**B** Using the same sample, does changing 95% confidence to 99% narrow the interval?

**C** A desired full interval width is 0.2 cm. What margin enters the n formula?

**D** At fixed confidence and σ, what happens to the margin when n is quadrupled?

**E** For a 95% one-sided lower bound, should the multiplier be 1.96 or 1.645?

For each answer, identify the rule or definition that justifies it.

# Concept check: answers and reasoning

**NAME THE QUANTITY, THEN APPLY THE APPROPRIATE RULE**

| | | |
|---|---|---|
| A | **No.** | 0.1 cm already measures the uncertainty in the sample mean. |
| B | **It widens.** | A higher confidence level uses a larger critical value. |
| C | **M = 0.1 cm.** | The margin is one half of the full width. |
| D | **The margin halves.** | M is proportional to $1/\sqrt{n}$ when z and σ are fixed. |
| E | **Use 1.645.** | A 95% one-sided bound places α = 0.05 in one tail. |

Choose the confidence statement and standard error before applying a formula.

# Lecture 13: key takeaways

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**ESTIMATE • QUANTIFY UNCERTAINTY • INTERPRET • PLAN**

| | |
|---|---|
| **Confidence interval** | Estimate a population mean using a centre and a margin. |
| **Standard error** | Describe uncertainty in the sample mean, not individual spread. |
| **Confidence level** | Describe long-run coverage of the interval-building method. |
| **Precision** | Higher confidence widens; a larger valid sample can narrow. |
| **One-sided bounds** | Use the appropriate direction and one-tail critical value. |

Final check: target mean, sampling conditions, standard error, and tail allocation.